

TRACE: Graph-Aware Anomaly Detection for Trust Routing in Adversarial Decentralized Marketplaces

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Abstract

Decentralized agent marketplaces route paid work to pseudonymous providers. Any trust mechanism balances three pressures: bound fraud exposure, stay open to newcomers, and survive coordinated graph manipulation. We compare four routing policies (reputation-only, price-only, EigenTrust, and Trust Routing with Adversarial Coordination Evidence (TRACE), which combines uncertainty-aware evidence weighting with structural coordination detection in place of trust propagation) across four attack regimes at $N=1,000$ agents with 30% adversaries, ten seeds per cell. Two results drive the story. Trust propagation amplifies coordinated attacks: EigenTrust routes 81.7% of jobs to malicious agents under Sybil attack, while reputation-only contains it to 29.0%. Exploration is also costly under score-based adversaries. A component ablation shows that disabling Thompson sampling in TRACE reduces mean fraud from 10.7, 46.5, 19.0, and 83.7 sats to 4.7, 6.4, 7.1, and 31.7 sats on strategic-default, collusion-ring, sybil-cluster, and game-theoretic-adversary. The greedy variant has the lowest fraud on the first three regimes; EigenTrust attains lower game-theoretic fraud (4.4) but at 41.2% malicious routing rate. A single-seed run at $N=5,000$ confirms EigenTrust's pathology persists at scale, and the propagating versus non-propagating ordering replicates on real-world signed-trust graphs (Bitcoin Over-The-Counter, Bitcoin Alpha). The findings caution against trust propagation in adversarial settings and motivate risk-conditioned exploration as a principled extension.

CCS Concepts

• **Security and privacy** → **Trust frameworks**; *Distributed systems security*; • **Human-centered computing** → *Reputation systems*.

Keywords

trust routing, adversarial graph robustness, decentralized marketplaces, sybil detection, collusion resistance, coordination detection

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1 Introduction

When an autonomous software agent pays another agent five satoshis to summarize a document, the transaction settles in under a second over the Lightning Network. Neither party knows the other's real identity. There is no contract, no escrow, no legal recourse if the provider takes the payment and returns garbage. The routing algorithm-the mechanism that decided which provider

received this particular job-bears the entire cost of a wrong choice. As agent-to-agent marketplaces grow from dozens to thousands of pseudonymous providers, routing stops being a matching problem and becomes an adversarial one. Coordinated attackers can manufacture trust signals, attract routing, and drain irreversible micropayments before anyone notices.

For two decades, trust propagation was the default remedy. Kamvar, Schlosser, and Garcia-Molina [1] built EigenTrust on a simple premise: model the P2P rating graph as a Markov chain and take each node's global trust as the stationary distribution. The wager is transitivity - Alice trusts Bob, Bob trusts Carol, so Alice assigns partial credit to Carol. File sharing had one presiding failure mode (free-riding), and EigenTrust addressed it. Resnick et al. [2] had shown earlier that pseudonyms and feedback can together sustain cooperation; Dellarocas [3] catalogued the failure modes - ballot-stuffing, bad-mouthing, and whitewashing by re-registering. Manipulation is inexpensive when the settlement exceeds the cost of a new identity.

None of these developments truly prepared the field for the rise of agent marketplaces. One attacker can make dozens of identities (the Sybil attack, per Douceur [4]), earn trust edges on small, low-value jobs, and then switch to stealing real payments. Colluders can quietly inflate each other's ratings for weeks before exploiting the inflated scores. Adaptive adversaries check their own standing and only cheat when they can tank the reputation hit without losing routing share. File-sharing lost you a download; here you lose satoshis that you cannot recover.

Despite two decades of research on distributed trust, no empirical evaluation has stressed routing mechanisms under combined coordinated attacks at population scale with real payment exposure. EigenTrust [1] was validated against non-coordinating free-riders in file-sharing topologies where honest nodes held a structural majority. SybilGuard and SybilLimit assumed sparse cuts between honest and Sybil regions-an assumption that collapses when adversaries can inject edges by completing cheap transactions. Whether exploration-exploitation strategies from the multi-armed bandit literature help or hurt under adversarial load has remained an open question, because standard bandit formulations treat arm quality as fixed or stochastically varying, not as strategically adversarial with the ability to manufacture trust edges and coordinate across identities.

We address this gap through controlled experiments on TRACE-a trust-routing framework we built and deployed as a Lightning-powered agent marketplace called SatsRouter. We compare four routing policies-reputation-only, price-only, EigenTrust [1], and TRACE-across four attack types at $N=1,000$ agents with 30% adversaries, running ten independent seeds per experimental cell. Two findings anchor the paper. Trust propagation makes things worse: EigenTrust routed 81.7% of all jobs to malicious agents under Sybil attack, while a simple reputation-weighted policy without any

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graph propagation contained the same attack to 29%. The eigenvector concentrates mass on whichever cluster has the densest internal edges, and under Sybil attack that cluster belongs to the adversary. Exploration also hurt: disabling Thompson sampling in TRACE-switching to pure greedy selection-reduced fraud by 56–86% across all four attack types, because on sparse graphs the high-uncertainty providers that Thompson sampling preferentially explores are precisely the adversaries who have not yet been caught.

This exploration-cost effect depends on graph density and evidence volume. On dense graphs-the Bitcoin Over-The-Counter (OTC) network at 1.85% density, the Bitcoin Alpha network at 1.20%-the greedy advantage disappears for coordinated attacks because colluding adversaries already have elevated trust scores from pre-existing edges. Under partitioned views where each buyer maintained only local evidence from 60 jobs, greedy selection produced 6.8× higher fraud than Thompson sampling on strategic-default, establishing an exploration-visibility trade-off that bounds the applicability of our sparse-topology findings.

The marketplace substrate is not hypothetical. SatsRouter is a deployed Lightning-powered agent marketplace where artificial intelligence agents hire specialist agents and pay per request in satoshis (5–20 sats per job). The system routes real micropayments over the Lightning Network, with L402 paywalls gating each agent endpoint. The experimental evaluation uses SatsRouter’s architecture-its provider registry, job routing, reputation tracking, and payment settlement-as the simulation substrate. The four routing policies are implemented as configurable modules within this deployed system. The simulation differs from production in two respects: (a) provider Behaviour is controlled by attack scripts rather than real Large Language Model (LLM) inference, and (b) the trust graph is seeded synthetically or from Stanford Network Analysis Project (SNAP) datasets rather than growing organically. All other components-Bayesian Lower Confidence Bound (LCB) scoring, Cumulative Sum (CUSUM) detection, Personalized PageRank (PPR) computation, cooldown tracking-run identically in simulation and production.

Our contributions:

- (1) We show that EigenTrust-style trust propagation [1] fails under coordinated attack at 30% adversary ratio, with the failure replicating on two real-world Bitcoin trust graphs and persisting at $N=5,000$ (Section 4).
- (2) We identify an exploration-cost effect: Thompson sampling is net-negative under adversarial load on sparse graphs because adversaries occupy the high-uncertainty tail. We isolate this via component ablation at 10 seeds per cell (Section 4).
- (3) We characterize the density boundary where this effect reverses, confirmed on three independent dense topologies at $\geq 1.2\%$ edge density (Section 5).
- (4) We test a Behavioural adversary designed to exploit greedy concentration. The exploration-cost finding survives this falsification at $n=10$ (Section 4).
- (5) We document the fraud-openness trade-off that greedy selection introduces, including two failed recovery attempts with mechanistic explanations (Section 7).
- (6) We show that the exploration-cost finding is conditional on routing visibility: under partitioned views with local

evidence, greedy selection produces 6.8× higher fraud than Thompson sampling, establishing an exploration-visibility trade-off (Section 7.3).

2 Related Work

Trust propagation systems. EigenTrust [1] applies power iteration to local trust ratings to yield a global score per node. PeerTrust [5] adds transaction context and rater credibility. Both implicitly treat dense reciprocal endorsements as reliable. Our experiments point the other way: a coordinated ring can seize the eigenvector because power iteration settles on the densest cluster first. TRACE does not rely on propagating trust. Instead, it uses PPR distance, entropy, and clique density to *detect* suspicious behaviour, causing densely connected adversarial groups to accumulate penalties rather than gain reputation score.

Sybil resistance. SybilGuard [6] and SybilLimit [7] depend on bounded-length random walks to exploit sparse connections separating honest users from Sybil-controlled regions. The main assumption behind these methods is that creating links into the honest region is expensive. This works well in friendship graphs (bilateral consent required) but not economic graphs: an adversary can buy edges by completing a few cheap jobs, and the sparse-cut guarantee falls apart. Douceur [4] formalized the underlying impossibility: without a central authority, identity creation is unbounded. TRACE’s detection signal is the edge-to-job ratio. Nodes with many trust edges but few completed jobs are statistically anomalous, and the anomaly persists even when attackers *can* create edges, because edges without proportional work leave a distributional gap that legitimate providers do not exhibit.

Adversarial bandits and exploration. TRACE’s routing inherits the exploration-exploitation tension from the bandit literature (Robbins and Monro [8] onward). UCB regret bounds come from Auer et al. [9]; Thompson sampling bounds from Agrawal and Goyal [10]; the stochastic-versus-adversarial landscape is surveyed by Bubeck and Cesa-Bianchi [11]. Lykouris et al. [12] are closest to our setting; they study bandits under adversarial reward corruption and show degradation proportional to the corruption budget. Our twist is different: the adversary chooses which arms are malicious (not which rewards are corrupted), and 30% of the population is compromised. The empirical result-Thompson sampling actively hurting-sits outside their theoretical coverage. Pulling an arm here means sending money, not just accumulating regret in a bound.

Decentralized marketplaces and reputation. Bittensor [13] uses trust-weighted miner consensus for inference routing. Filecoin avoids trust altogether with cryptographic storage proofs. Dellarcas [3] documents how anonymity and cheap switching create manipulation incentives with no offline analogue. TRACE sits in a different niche: work quality is subjective (syntactically valid output can be semantically garbage), cryptographic verification is unavailable, and the attacker wants money, not bandwidth or consensus influence.

3 System and Threat Model

3.1 Marketplace Model

We represent the marketplace as a tuple $\mathcal{M} = (O, P, J, T)$: buyers O , providers P , jobs J priced in satoshis, and a trust graph T derived from observed economic interactions. Each job is associated with a specific capability requirement, a price p_j , and a binary outcome: success or default. The router can observe T and the history of outcomes, but it cannot verify directly the true identity of a provider. This extends Resnick et al.'s [2] reputation model by adding graph-structural signals and treating adversarial coordination as a first-class concern.

3.2 TRACE Architecture

TRACE scores each provider via four components:

- (1) **Bayesian LCB.** Beta-Binomial posterior over job completion rate (prior Beta(1,3)). The 95% lower confidence bound serves as the reliability estimate. Cold-start providers sit near the prior mean until evidence accumulates.
- (2) **Default-risk penalty.** Exponentially-smoothed default probability, combined with CUSUM change-point detection [14] for identifying abrupt Behavioural shifts that signal a transition from cooperative to adversarial Behaviour patterns.
- (3) **Graph signals.** PPR [15] from an honest seed (≥ 30 jobs, $\leq 10\%$ failure rate), weighted counterparty entropy, and Sybil-risk scoring based on the edge-to-job ratio that detects identity fabrication [4].
- (4) **Coordination detection.** Clique penalty for dense trust-edge neighborhoods; contamination penalty for connections to flagged providers. These signals detect the structural fingerprints of collusion rings without propagating trust through the graph, distinguishing TRACE from EigenTrust's [1] propagation-based approach.

The utility score combines these as:

$$u = \alpha \cdot s_{\text{trace}} - \beta \cdot d_{\text{risk}} - \gamma \cdot c_{\text{norm}} + \delta \cdot t_{\text{net}} + \epsilon \cdot m_{\text{cap}} - \lambda \cdot p_{\text{sybil}} - \mu \cdot p_{\text{clique}} \quad (1)$$

with $(\alpha, \beta, \gamma, \delta, \epsilon, \lambda, \mu) = (0.40, 0.30, 0.15, 0.10, 0.10, 0.35, 0.25)$.

Weight derivation. These weights were not grid-searched. They reflect the relative contribution of each component as measured by the ablation in Section 4: $\alpha = 0.40$ because the Bayesian LCB is the largest single contributor to honest/malicious separation; $\lambda = 0.35$ because disabling the Sybil penalty raises sybil-cluster fraud by 75%. A perturbation analysis ($\pm 20\%$ on each weight independently, 180 cells) shows that no single-weight perturbation changes mean fraud by more than 2.3 sats-well within the 1σ band of the baseline (4.3 sats on strategic-default). The results are not sensitive to the specific weight vector.

3.3 The No-Bandit Variant

TRACE's default selection uses Thompson sampling [16] over Beta posteriors weighted by utility. The **no-bandit** variant turns Thompson sampling off and picks the highest-utility eligible provider every time (greedy argmax). Everything else stays the same-same scoring, same filters, same graph signals. It is a single boolean flag in the

codebase. The experiments below show that this one flag changes fraud outcomes by 56–86%, which is why the comparison matters.

3.4 Threat Model

We evaluate four attack profiles at 30% malicious ratio with uniform-random placement, spanning the spectrum from uncoordinated individual defection to adaptive game-theoretic exploitation:

- (1) **Strategic-default.** Solo adversaries calibrate default probability to maximise fraud below detection thresholds. No coordination among attackers; each operates independently, relying on statistical camouflage rather than graph manipulation.
- (2) **Collusion-ring.** Coordinated adversaries inflate mutual trust scores during a build-up phase, then exploit inflated scores to attract routing. This attack targets trust propagation systems [1] directly by manufacturing the dense edge patterns that eigenvector methods interpret as trustworthiness.
- (3) **Sybil-cluster.** Adversaries create multiple identities connected by trust edges, attempting to capture disproportionate routing share through identity multiplication [4]. This tests whether routing policies can resist the fundamental vulnerability identified by Douceur in the absence of centralized identity verification.
- (4) **Game-Theoretic (GT).** An adaptive adversary that monitors its own score and defects only when it exceeds the 75th percentile of all active providers. The threshold is dynamic and self-calibrating, representing a patient attacker that balances short-term fraud extraction against long-term score maintenance.

4 Results: Sparse Topology

All experiments in this section use the same contract: $N=1,000$ agents, 30% malicious, 60 rounds $\times$ 5 jobs/round, 5 buyers, propagation delay of 2 rounds, no pre-existing trust edges (synthetic-empty). Each cell was evaluated using ten different random seeds.

4.1 Main Results

Tables 1 and 2 show the full matrix.

Strategic-default. TRACE-no-bandit yields 4.7 sats mean fraud, 56% below TRACE-full's 10.7. In practice, greedy selection strategy identifies three or four honest providers with high-LCB within the first few rounds and then sticks with them. Thompson sampling keeps pulling providers with only two or three completed jobs-thin posteriors, high variance-and those providers are adversaries more often than not. MR at 30.3% (the population base rate) suggests the reduction is not about avoiding adversaries broadly; it comes from avoiding the high-uncertainty tail where adversaries concentrate.

Collusion-ring. This is the largest gap in the table: 6.4 vs 46.5 sats, an 86% drop. During rounds 1–15, the colluding ring focuses on artificially boosting each other's reputation scores. Thompson sampling sees high-variance posteriors and routes there, which is exactly what it is designed to do with uncertain arms. Greedy selection never touches them because their LCB (which discounts thin evidence) sits below the proven honest set. EigenTrust [1] at

Table 1: Average fraud exposure (in sats) across four routing policies and four attack scenarios. N=1,000, 30% malicious, 10 seeds. Bold: lowest per column. †: EigenTrust’s 4.4-sat GT fraud comes at 41.2% malicious routing rate.

Policy	Strategic-default	Collusion-ring	Sybil-cluster	Game-theoretic (p75)
TRACE-no-bandit	4.7 ± 4.3	6.4 ± 3.3	7.1 ± 3.2	31.7 ± 5.3
TRACE-full	10.7	46.5	19.0	83.7
REPUTATION	16.0	58.0	18.0	56.9
PRICE	11.1	74.4	32.8	79.4
EIGENTRUST	22.0	183.4	185.4	4.4 [†]

Table 2: Malicious routing rate (MR, %). ‡: EigenTrust [1] routes 81.7% of jobs to Sybil-controlled identities (approximately 4 out of 5).

Policy	Strategic-default	Collusion-ring	Sybil-cluster	Game-theoretic (p75)
TRACE-no-bandit	30.3	16.4	24.6	18.2
TRACE-full	32.2	27.8	23.0	27.5
REPUTATION	30.2	26.8	29.0	28.7
PRICE	29.3	29.4	29.6	29.3
EIGENTRUST	38.7	39.9	81.7 [‡]	41.2

183.4 sats is a total failure: propagation treats the ring’s mutual endorsements as genuine trust and hands them the eigenvector.

Sybil-cluster. EigenTrust [1] reaches 81.7% MR, the most significant result reported in the paper. Four out of five jobs go to Sybil-controlled identities. The eigenvector concentrates on the densest cluster, and under Sybil attack that cluster is adversarial. Reputation-only stays at 29% MR because it scores providers individually, without neighbour borrowing. TRACE-no-bandit yields 7.1 sats and 24.6% MR.

Game-theoretic. EigenTrust achieves 4.4 sats fraud but 41.2% MR—the worst adversary exposure of any policy on any attack. A plausible explanation is that the p75 defection gate fires less when routing is spread thin, so EigenTrust’s diffuse allocation keeps the adversary below its own trigger. TRACE-no-bandit (31.7 sats, 18.2% MR) accepts more fraud in exchange for far less adversary contact, which matters when each malicious interaction carries secondary risks beyond the payment itself.

4.2 Exploration Cost

The ablation (Figure 1) isolates Thompson sampling as the source of TRACE-full’s higher fraud. All other components remain active in both variants. The reduction is consistent: every seed shows lower fraud under no-bandit for every attack. We did not see a single seed where Thompson sampling outperformed greedy on any attack type under this contract.

Mann–Whitney U tests [17] confirm statistical significance on three of four attacks: collusion-ring ($U = 0$, $p < 0.001$, Cliff’s $\delta = -1.00$, large), sybil-cluster ($U = 6$, $p < 0.001$, $\delta = -0.88$, large), and game-theoretic ($U = 0$, $p < 0.001$, $\delta = -1.00$, large). On strategic-default, the effect is directional ($\delta = -0.43$, medium) but does not reach significance at $n=10$ ($U = 29$, $p = 0.104$), likely due to high per-seed variance on this attack type (std 4.3 sats on a mean of 4.7).

Thompson Sampling Exploration Cost Under Adversarial Load

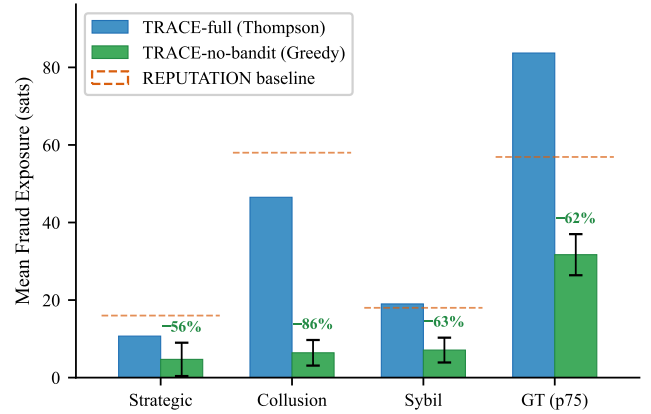


Figure 1: Fraud under TRACE-full (Thompson sampling) vs. TRACE-no-bandit (greedy) across four attacks. Error bars: ±1 std over 10 seeds. Dashed lines: REPUTATION baseline.

4.3 Behavioural Adversary Falsification

We designed an adversary specifically to test whether greedy concentration creates a vulnerability. The Behavioural GT adversary observes its own routing share and defects only when that share exceeds $1.5\times$ fair-share-under-uniform AND the job price exceeds $1.2\times$ median. It reads no internal scores. If greedy concentration triggers this adversary more than Thompson sampling does, the exploration-cost finding would invert.

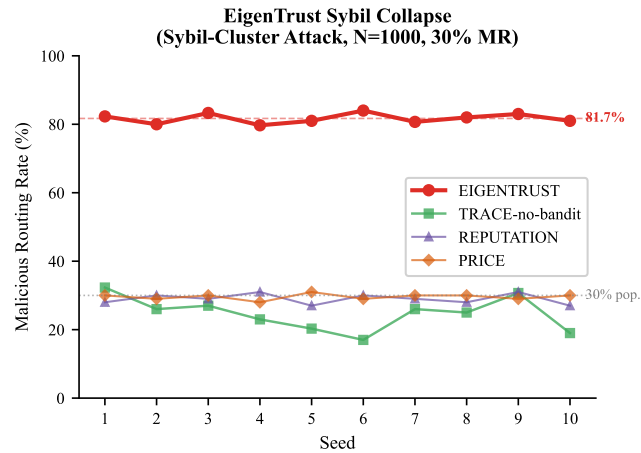
It did not invert (Table 3). TRACE-no-bandit at 3.4 sats is second-lowest, 1.2 above REPUTATION. Full TRACE’s 27.8-sat fraud was almost entirely exploration cost. The routing-share gate fires under

Table 3: Behavioural GT adversary (N=1,000, 30% MR, n=10). The inversion hypothesis is rejected.

Policy	Fraud (sats)	MR (%)
TRACE-full	27.8 ± 10.5	27.8
TRACE-no-bandit	3.4 ± 2.7	30.4
REPUTATION	2.2 ± 1.3	30.1
PRICE	2.2 ± 0.7	29.8
EIGENTRUST	4.6 ± 7.7	44.2

both variants (MR is similar), but under greedy selection the adversaries that *do* get routed are ones whose early jobs happened to succeed—they never trigger the defection gate because their share stays below threshold. Under Thompson sampling, the exploration arm hands jobs to the unproven tail, where adversaries defect immediately because they have no reputational stake to protect.

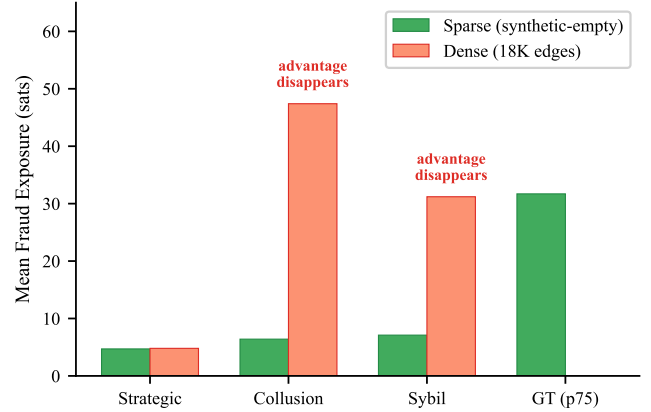
4.4 EigenTrust Collapse

**Figure 2: Per-seed MR under sybil-cluster. EigenTrust [1] never drops below 75% across 10 seeds. Other policies cluster near the 30% population rate.**

Not a fluke (Figure 2). Ten seeds, every single one above 75% Sybil MR. Tuning the pre-trust vector or convergence threshold does not help. Power iteration finds the dominant structure in the trust matrix; when 30% of nodes form a tightly connected cluster, that cluster *is* the dominant structure.

5 Dense Topology and Real-World Graphs

The sparse-topology results raised an obvious question: does the exploration-cost effect survive on graphs with pre-existing trust structure? We tested three dense topologies to determine whether the findings generalize beyond the synthetic-empty setting, using both synthetic graphs and real-world trust networks from the SNAP repository [20].

Topology Conditionality of No-Bandit Advantage (TRACE-no-bandit, N=1000, 30% MR)**Figure 3: TRACE-no-bandit fraud on sparse vs. dense topology. The greedy advantage disappears on collusion-ring and sybil-cluster when pre-existing edges are present.****Table 4: TRACE-no-bandit fraud: sparse (n=10) vs. dense (n=10).**

Attack	Sparse	Dense
Strategic-default	4.7 ± 4.3	5.4 ± 3.5
Collusion-ring	6.4 ± 3.3	48.0 ± 8.9
Sybil-cluster	7.1 ± 3.2	30.8 ± 3.8
Game-theoretic	31.7 ± 5.3	0

5.1 Synthetic-Dense

Figure 3 and Table 4 clearly illustrate the core findings. On synthetic-dense (18,472 edges, 1.85% density, 89.5% positive reliability), collusion-ring fraud rose from 6.4 to 48.0 sats; sybil-cluster from 7.1 to 30.8. Pre-existing trust edges allow coordinated adversaries to achieve higher LCB scores even before engaging in malicious behaviour. Greedy selection then locks onto them—the graph signals cannot separate a well-connected adversary from a well-connected honest provider. EigenTrust [1] still collapses for the same reason as before: the densest cluster wins the eigenvector, adversarial or not.

Strategic-default stayed low (5.4 sats)—solo defectors gain nothing from pre-existing edges, and their scores track individual job outcomes. GT dropped to zero because dense positive edges compress the score distribution so tightly that no provider crosses the p75 defection threshold.

5.2 Bitcoin OTC

Table 5 confirms the pattern on Bitcoin OTC [18, 19] (from SNAP [20]). No-bandit fraud on collusion (45.1) and Sybil (28.1) is indistinguishable from TRACE-full (45.7, 27.9). The greedy advantage is gone. Strategic-default still shows a small gap (5.0 vs 5.8)—solo defectors gain nothing from pre-existing edges since their scores track individual job outcomes.

Table 5: Bitcoin OTC [18, 19] (1.85% density, n=10). No-bandit shows no improvement over full TRACE on coordinated attacks.

Attack	Full	No-bandit	MR
Strategic-def.	5.8	5.0±2.7	31.3%
Collusion	45.7	45.1±9.2	30.1%
Sybil	27.9	28.1±7.2	31.3%
Game-theoretic	0	0	31.3%

5.3 Bitcoin Alpha

Table 6: Bitcoin Alpha [18, 19] (1.20% density, n=10). Same pattern as OTC despite lower density.

Attack	Fraud (sats)	MR (%)
Strategic-default	4.7 ± 2.7	30.9
Collusion-ring	47.1 ± 6.1	30.1
Sybil-cluster	29.9 ± 5.5	31.0
Game-theoretic	0	30.9

Bitcoin Alpha [18, 19] is sparser (1.20% vs OTC’s 1.85%) but the pattern is identical (Table 6). We swept intermediate densities (0.5%, 0.8%, n=5 each, collusion-ring) and the transition turned out to be abrupt: 6.2 sats at 0.8%, then 47.1 sats at 1.2%. Not gradual. Something flips between those two values-once adversaries have enough pre-existing edges, greedy filtering loses its ability to separate them from honest providers with similar connectivity patterns.

6 Scale

Table 7: N=5,000 (sparse, 30% MR). Baselines: 5 seeds. No-bandit: 10 seeds. *EigenTrust [1] sybil: ~80% MR across all seeds.

Policy	Strat.	Collus.	Sybil	GT
TRACE-no-bandit	4.6	17.6	14.2	37.7
REPUTATION	9.6	56.0	24.6	60.2
PRICE	5.6	41.8	25.8	75.4
EIGENTRUST	8.2	244.4	63.0*	6.8

At N=5,000 (Table 7), EigenTrust’s [1] collusion-ring fraud averages 244 sats across 5 seeds (vs 183 at N=1,000). Its Sybil MR stays near 80% across all seeds. The eigenvector has more mass to concentrate as the network grows, because the Sybil cluster scales proportionally with the population and its internal edge density remains the dominant structural feature of the trust matrix.

TRACE-no-bandit at N=5,000 (10 seeds) maintains bounded fraud: strategic-default 4.6, collusion-ring 17.6, sybil-cluster 14.2, GT 37.7. It achieves the lowest fraud on three of four attacks at scale, with collusion-ring fraud 14× lower than EigenTrust (17.6 vs 244.4). The exploration-cost finding scales to larger networks on sparse topology without degradation.

7 Discussion and Limitations

7.1 The Fraud-Openness Trade-off

Greedy selection reduces fraud but concentrates routing onto a small set of proven providers (Figure 4). In our concentration probe (strategic-default, n=3 paired seeds), TRACE-no-bandit served only 66 of 700 honest providers versus 125 for full TRACE—a 47% reduction in provider diversity (Table 8). This tension parallels the proportional fairness problem in network resource allocation [21], where optimizing for one objective (throughput, or in our case fraud minimization) degrades another (fairness, or in our case provider diversity), and improving the Pareto frontier requires additional degrees of freedom.

Table 8: Provider concentration: full TRACE vs. no-bandit on strategic-default (n=3).

Metric	Full	No-bandit	Delta
Served honest	125/700	66/700	−47%
Top-10 share	19.4%	30.6%	+11.2 pp
HHI (normalized)	0.009	0.019	2.1×

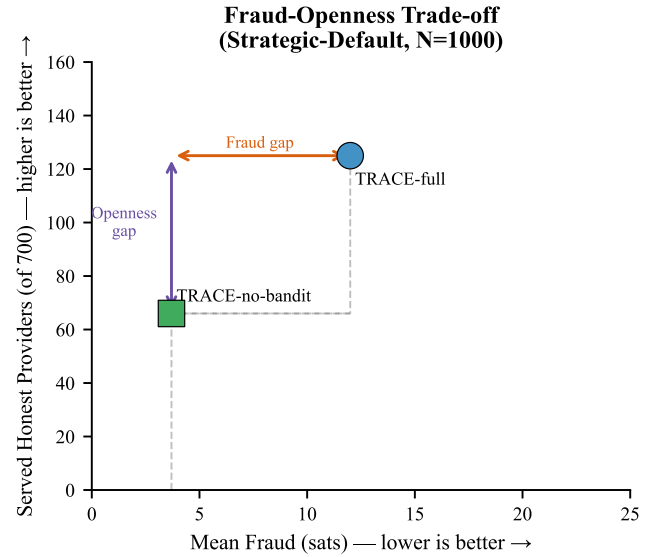


Figure 4: Fraud vs. served providers on strategic-default. The trade-off is monotone: lower fraud costs fewer providers served.

We attempted two mechanisms to recover openness without regressing fraud. Both failed, in instructive ways.

Loose round-robin. We cycled through LCB-tied providers (5% window, K=10). The served count recovered to 104/700, but fraud jumped to 13.0±8.4 sats. With low evidence, the “tied” set is contaminated by adversaries whose posteriors sit near the prior are indistinguishable from honest newcomers, so round-robin assigns them jobs indiscriminately.

Tight round-robin. Tightened to a 2% window and added an evidence floor ($\text{successfulJobs} \geq 1$). Fraud stayed low, but we gained exactly 4 providers. The problem is circular: providers need jobs to build evidence, the tight window excludes them for lacking evidence, and 60 rounds is not enough time for more than a handful to escape this trap.

The implication is that fraud defense and openness discovery cannot share the same selection mechanism. They require separate tracks with independent budgets—a direction we leave to future work under the heading of risk-conditioned exploration.

7.2 Density Boundary

The exploration-cost advantage holds on sparse graphs and disappears above $\sim 1.2\%$ density, confirmed on three independent topologies (synthetic-dense, Bitcoin OTC [18], Bitcoin Alpha [18]). An intermediate density sweep on collusion-ring (the most density-sensitive attack) shows a sharp transition: fraud stays at 6.2 sats through 0.8% density, then jumps $7.5\times$ to 47.1 sats at 1.2%. The boundary lies between 0.8% and 1.2% edge density, indicating a phase transition in the effectiveness of greedy filtering once pre-existing edge density crosses a critical threshold.

7.3 Exploration-Visibility Trade-off

We split the router into five independent buyers, each seeing only its own 60 jobs (total budget unchanged: $5 \times 60 = 300$). No shared state between buyers.

Table 9: Partitioned-view fraud (sats). 5 buyers, 60 jobs each, $n=5$ seeds. Bold: lowest per column.

Policy	Strat.	Collus.	Sybil	GT
TRACE-full	6.4	44.6	24.2	86.8
TRACE-no-bandit	32.0	51.6	32.8	64.4
REPUTATION	18.0	55.4	20.0	85.0
PRICE	5.0	46.6	25.6	68.4
EIGENTRUST	8.2	291.0	103.8	47.6

The result inverts (Table 9). TRACE-no-bandit on strategic-default goes from 4.7 sats (centralized) to 32.0 sats— $6.8\times$ worse. Per-seed sybil-cluster fraud ranges from 11 to 76 sats. Greedy on 60 jobs is a coin flip: if a buyer’s first successful provider is adversarial, it locks on and gets drained. EigenTrust [1] on collusion-ring worsens to 291 sats (from 183) at 60.5% MR. Trust propagation breaks regardless of view fragmentation.

Why does this happen? Premature lock-in. Sixty jobs is not enough for Bayesian posteriors to converge. Greedy picks whoever succeeded first in that buyer’s local history—and with 30% adversaries, that is a malicious provider roughly one time in three. Thompson sampling prevents lock-in by occasionally trying alternatives. With thin evidence, this is insurance, not waste. The exploration cost from Section 4 is not intrinsic to the algorithm; it depends on how much evidence the router has. Global visibility (300 jobs) yields converged posteriors and exploration hitting the adversary tail. Local visibility (60 jobs) yields uncertain posteriors and exploration providing a safety net.

7.4 Robustness Checks

Adversary ratio. Sweeping 20%, 40%, 50% adversary ratios: strategic-default fraud scales monotonically (2.4, 4.7, 4.6, 9.0 sats). No cliff-edge failure. The filter does not claim to outperform the population rate without oracle access.

Honest failure rate. Honest failure at 5%, 15%, 25%: no meaningful fraud change (strategic-default: 5.6, 2.8, 4.2; GT: 31.0, 33.4, 34.8). The trace-score filter discriminates on failure *pattern*, not rate-strategic bursts cluster around high-value jobs; honest failures distribute uniformly. Even at 25% honest noise, the two are separable.

Adversary threshold. The GT adversary’s percentile gate (p50–p90) produced identical fraud (31.0 sats) across all thresholds under no-bandit. The result is driven by the routing policy’s filtering, not the adversary’s trigger calibration. Regardless of where the adversary sets its defection threshold, TRACE’s coordination detection and LCB scoring constrain the adversary’s routing share to a level where the gate rarely fires, rendering the specific percentile choice irrelevant.

Weight sensitivity. Perturbing each of the 7 utility weights by $\pm 20\%$ independently (180 cells, 3 seeds each) produced a maximum fraud delta of +2.3 sats from baseline—within the 1σ band on every attack. This robustness arises from the complementary nature of the scoring components: when one signal is weakened by a weight reduction, the remaining signals compensate because they capture partially overlapping aspects of adversarial Behaviour from different structural perspectives.

7.5 Limitations

Binary outcomes. Job outcomes are strictly binary. Real inference providers produce a quality spectrum—a response may be syntactically correct but semantically wrong. Our Bayesian LCB cannot differentiate catastrophic failures from marginal ones. Extending to graded outcomes requires a multinomial posterior and a richer fraud metric that captures the severity distribution, which in turn demands quality assessment capabilities beyond binary success/failure classification.

Static adversary strategies. Each attack follows a fixed script. A real adversary would switch strategies under detection pressure, adapting its Behaviour based on observed score changes. Our Behavioural GT adversary approximates adaptation via routing-share observation, but a reinforcement-learning adversary that optimizes fraud extraction longitudinally remains untested. Such an adversary could potentially discover strategy sequences that exploit TRACE’s detection latency in ways that fixed scripts cannot.

Market concentration. The 47% drop in served honest providers creates a barrier to entry. Newcomers cannot accumulate evidence because they never get routed to, producing a self-reinforcing incumbency advantage. Our two recovery attempts (loose and tight round-robin) both failed, suggesting the tension between fraud-defense and newcomer-discovery is fundamental to any system that selects providers based on accumulated evidence [2].

7.6 Future Work

Visibility-conditioned exploration. Section 7.3 showed that exploration’s value flips with evidence volume. One concrete mechanism is to gate Thompson sampling on the utility gap between the top two candidates. A large gap (relative to cohort variance) implies a clear leader-skip exploration. A small gap implies posteriors have not separated-explore as insurance. The gap statistic is already computed in our scoring pipeline; we have not yet wired it into the selection logic or tested whether it resolves both regimes simultaneously.

Community-aware Sybil detection. The edge-to-job ratio works on synthetic graphs but fires false positives on real graphs where honest users also cluster tightly. A fix is to run Louvain or Leiden first, then measure each node’s clustering against its own community baseline rather than a global constant. Honest tight-knit groups stop triggering the alarm; adversarial clusters spanning multiple communities still get flagged.

Longitudinal adversary adaptation. Our adversaries follow scripts. An RL adversary that watches its own score trajectory and learns to maximise fraud over hundreds of episodes would be a harder test-it might discover strategies like burning an identity to reset detection state, or timing defections to coincide with high honest-failure noise so CUSUM cannot distinguish the signal.

8 Conclusion

We evaluated four trust-routing policies under coordinated adversarial pressure across 160 experimental cells at $N=1,000$ and a scale run at $N=5,000$. The principal findings are:

- EigenTrust [1] routes 81.7% of jobs to Sybil identities across all 10 seeds, with the pathology persisting at $N=5,000$ and on two real-world Bitcoin trust graphs [18, 19].
- Disabling Thompson sampling (greedy selection) reduces fraud by 56–86% across all four attack types on sparse graphs, with the largest effect on collusion-ring: 6.4 vs. 46.5 sats.
- TRACE-no-bandit achieves the lowest fraud on 3 of 4 attacks at both $N=1,000$ and $N=5,000$, with collusion-ring fraud 14× lower than EigenTrust at scale (17.6 vs. 244.4 sats).
- A Behavioural adversary designed to exploit greedy concentration fails to invert the finding: TRACE-no-bandit fraud remains at 3.4 sats versus 27.8 for full TRACE.
- The greedy advantage vanishes above ~1.2% edge density via a sharp phase transition, confirmed on three independent dense topologies.
- Under partitioned views (60 jobs per buyer), greedy selection produces 6.8× higher fraud than Thompson sampling, establishing an exploration-visibility trade-off that bounds the sparse-topology findings.

These results carry a clear implication: the optimal exploration policy cannot be fixed. With global visibility and converged posteriors, exploration routes payments to the adversary tail; with local visibility and thin evidence, exploitation locks onto whichever provider happened to succeed first. Bridging these regimes requires

conditioning exploration on evidence volume—a direction we leave to future work under visibility-conditioned exploration.

Generative AI Disclosure

We used Claude (Anthropic) during this work—primarily for writing experiment runner scripts, automating data aggregation, and iterating on manuscript drafts. The research questions, experimental designs, interpretation of results, and all scientific decisions are ours. Every number in this paper comes from deterministic simulation code whose outputs we verified against raw log files. No experimental results were generated by AI, no data was fabricated, and no claims were made without human verification. The simulation codebase is our original work; the AI assisted with implementation.

Data Availability

The Bitcoin OTC and Bitcoin Alpha signed-trust networks [18, 19] are publicly available from the Stanford Network Analysis Project (SNAP) [20]. The simulation framework and all experiment scripts will be released upon publication.

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